

Two Vectors of Geopolitical Dependency in Large Language Models: Implications for the Global South and Directions for Inclusive AI Capacity

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Abstract

Large language models (LLMs) are increasingly embedded in critical applications, including health, finance, and education systems, in countries that depend on a small number of frontier providers. We argue that this dependency exposes the Global South to two distinct and compounding mechanisms of geopolitical control. The first is downstream: access to a model can be withdrawn unilaterally by the jurisdiction in which its developer is incorporated, independent of the needs of the countries that rely on it. The second is upstream: LLMs encode the political ideologies and epistemologies of the cultures and corpora that produce them, and a small number of countries currently dominate both training-data production and model distribution. We situate these mechanisms within the material infrastructure of overseas digital investment programmes, which often determine which frontier providers a country depends on in the first place. We outline actionable, low-resource strategies through which the Global South can reduce this dual exposure: locally grounded evaluation resources, adaptation of open-weight models under constrained compute, and mentorship pipelines that build regional capacity rather than deepening dependency on external providers.

1 Introduction

Countries and companies across the Global South are building health, finance, agricultural advisory, and education applications on top of a small number of frontier large language models (LLMs). This concentration creates a form of dependency that is rarely examined alongside the more familiar concerns of the inclusive machine learning (ML) community, namely data scarcity, mentorship, and low-resource adaptation. We argue that dependency on frontier LLMs is itself a low-resource problem, in the sense that countries and communities without sovereign compute or training-data capacity are structurally exposed to decisions made elsewhere, and that this exposure operates through two distinct mechanisms.

The first mechanism is downstream and concerns access. On 12 June 2026, the United States government issued an ex-

port control directive suspending access to two frontier systems, Claude Fable 5 and Claude Mythos 5, for foreign nationals; the developer, Anthropic, disabled the models for affected customers globally as a result, before access was restored on 1 July 2026 [Anthropic, 2026]. This episode illustrates a general point: frontier model access is contingent on the jurisdiction in which a developer is incorporated, not on the operational needs of the countries that depend on the model. For a country that has built critical infrastructure on top of a single foreign LLM, sudden withdrawal of access is a structural shock rather than an inconvenience.

The second mechanism is upstream and concerns training data. LLMs encode the political ideologies and epistemologies of the cultures and corpora that produce them [Waight *et al.*, 2026], and the countries that currently dominate both training-data production and model distribution are concentrated in the Global North and in China. Where a national information ecosystem is itself shaped by externally produced or state-coordinated media, adopting a foreign LLM compounds rather than corrects this dependency [Waight *et al.*, 2026]. Values embedded in ML research and in the resulting systems are not neutral [Birhane *et al.*, 2022], and the risk for underrepresented regions is not merely inaccurate output but the gradual displacement of local epistemic traditions by frameworks embedded, often invisibly, in training data produced elsewhere [Moiloa, 2026].

These two mechanisms are not new observations in isolation, but their interaction, and their relevance to the practical work of inclusive ML, has received comparatively little attention. This paper makes three contributions. First, we describe the two mechanisms and how they compound for countries with limited sovereign AI capacity. Second, we situate them within the material infrastructure through which digital dependency is often established in the first place. Third, and most directly relevant to this workshop, we outline concrete, low-resource strategies that the inclusive ML community, including student and early-career researchers, can pursue to reduce this dual exposure.

2 Two Mechanisms of Dependency

2.1 Downstream: Access Control and Vendor Lock-in

Frontier LLMs are typically served through a small number of application programming interfaces controlled by a handful of companies headquartered in a small number of jurisdictions. When a government restricts export of a model, as in the June 2026 case, the restriction propagates automatically to every downstream deployment, regardless of the deploying country’s own policy preferences or operational needs [Anthropic, 2026]. Countries that have built critical applications, such as clinical decision support, credit scoring, or agricultural advisory tools, on top of a single provider inherit this fragility without having had any voice in the decision that created it. This resembles the vendor lock-in long documented in enterprise cloud computing, but extended to a general-purpose cognitive technology on which entire public services may come to depend. Nations with less diversified digital infrastructure, which are disproportionately concentrated in the Global South, are likely to be less able to absorb such shocks than economies with domestic alternatives already in place, though we are not aware of published, quantitative evidence directly measuring this differential resilience.

2.2 Upstream: Training Data and Epistemic Displacement

The second mechanism operates independently of access. Waight et al. provide large-scale empirical evidence that government control of domestic media propagates, through training data, into the political orientation of LLMs, with a measurable effect in languages associated with lower press freedom [Waight et al., 2026]. This finding generalizes a longer-standing concern in the fairness and accountability literature that the values encoded in ML research and systems reflect the priorities and assumptions of the communities that build them [Birhane et al., 2022].

For communities whose epistemic traditions are underrepresented in the pretraining corpora of dominant models, for example many Indigenous and Afrocentric knowledge systems, the risk is not confined to isolated factual errors but extends to the gradual displacement of locally grounded ways of reasoning about a problem [Moiloa, 2026].

3 The Material Substrate of Dependency

Neither mechanism operates in a vacuum. The physical infrastructure, including fibre optic networks, data centres, and mobile broadband, through which a country accesses any LLM at all is frequently financed through large overseas digital investment programmes, of which China’s Digital Silk Road is the most extensively documented example [Hillman, 2021]. A country whose civil service or public information systems run on infrastructure financed through such a programme may inherit a physical dependency and, given the findings on training-data provenance described above, a compounding epistemic one. Crawford’s observation that AI systems encode the politics of the material and institutional systems in which they are embedded [Crawford, 2021] applies

	Upstream	Downstream
Mechanism	Training data composition	Export controls, licensing terms
Controlled by	Corpus curators, state media apparatus	Model operator’s home jurisdiction
Primary harm	Epistemic displacement	Operational disruption, vendor lock-in
Most exposed	Underrepresented language and knowledge communities	Nations with critical infrastructure on a single foreign model

Table 1: Two mechanisms of geopolitical dependency in large language models and their compounded effect on the Global South.

directly here: hosting model weights locally does not by itself confer sovereignty, since the training corpus and the terms of continued access to updates and support may still originate elsewhere. The two mechanisms in Section 2 and the material substrate described here jointly determine which countries are most exposed, as summarized in Table 1.

History offers a relevant caution. The telegraph was, in its time, a transformative information technology that nonetheless became an instrument of imperial control, with access and content shaped by the strategic interests of the states that operated the cable networks [Headrick, 1991].

Without coordinated attention to both training-data provenance and access continuity, LLMs risk entrenching a comparable form of dependency rather than narrowing the gap they are frequently presented as closing.

4 Toward Actionable Strategies for Inclusive Machine Learning

The mechanisms described above are structural, but the inclusive ML community, including students, postdocs, and community-led initiatives with limited compute and funding, is not without options. We outline three directions consistent with this workshop’s focus on inclusive ML, mentorship, and low-resource learning, and we frame each as a means of reducing exposure to the two mechanisms above rather than as a call to withhold frontier model access, which remains a genuine driver of innovation, productivity, and job creation that economies in the Global South need.

Locally grounded evaluation resources. Benchmarks constructed primarily from Global North data and assumptions are poorly positioned to detect the epistemic displacement described in Section 2.2. Community-built, low-resource benchmarks that centre local languages, health communication norms, and knowledge traditions provide a mechanism for detecting and documenting where a given model’s outputs diverge from locally grounded expectations, and they can be built incrementally by small teams without frontier compute budgets. Such efforts are most effective when developed in partnership with the communities whose knowledge

they aim to represent, rather than constructed unilaterally by external researchers.

Low-resource adaptation of open-weight models. Fine-tuning or lightweight adaptation of smaller, open-weight models on regionally curated data reduces, though does not eliminate, both mechanisms of dependency: it diversifies the set of models a country or institution relies upon, and it allows underrepresented epistemic content to be reintroduced downstream of a foreign pretraining corpus. This strategy is deliberately positioned as a complement to, rather than a substitute for, frontier model access, since low-resource adaptation alone cannot match the capability of frontier systems for many tasks.

Mentorship and capacity-building pipelines. Technical mitigations are of limited value without a regional talent pipeline able to build, evaluate, and maintain them. Structured mentorship programmes that connect early-career researchers in under-resourced regions with experienced practitioners, together with community-facing AI literacy initiatives aimed at students and educators, build the human capacity needed to sustain locally grounded evaluation and adaptation efforts over time. One example, details of which are withheld here to preserve the anonymity of this submission, is a volunteer-led programme delivering AI education materials to schoolchildren and educators in under-served communities in partnership with local community organizations. We note this as an illustration of the kind of low-cost, community-anchored initiative that this strategy entails, rather than as a claim of novelty.

Multilateral governance advocacy. Finally, researchers and research communities concerned with global equity can advocate for LLM governance frameworks that treat training-data provenance and access continuity as a joint problem rather than addressing either in isolation, drawing on the precedent of multilateral frameworks for satellite spectrum and international telecommunications. Absent such frameworks, the countries and communities least able to build frontier alternatives of their own will continue to bear the largest share of both forms of dependency described in this paper.

5 Conclusion

Large language models are becoming instruments of statecraft through two distinct mechanisms: downstream control over access, and upstream shaping of training data. Both are compounded by the material infrastructure through which many countries first gain access to digital and AI systems at all. We do not argue that frontier LLM access should be withheld from the Global South; rather, we argue for *prudence and reciprocity* in how that access is extended, and for a low-resource, community-anchored research agenda, spanning locally grounded evaluation, open-weight adaptation, and mentorship, through which the inclusive ML community can reduce the Global South's dual exposure to these mechanisms while frontier access questions are resolved at the governance level.

Ethical Statement

This paper discusses structural risks of unequal access to and representation within large language models, with a focus on their disproportionate impact on the Global South. We are not aware of any ethical concerns specific to this analysis beyond those already implied by the subject matter, namely global inequality in AI access and representation, which the paper's argument is intended to address rather than exacerbate.

References

- [Anthropic, 2026] Anthropic. Update on claude fable 5 and claude mythos 5 access, 2026.
- [Birhane *et al.*, 2022] Abeba Birhane, Pratyusha Kalluri, Dallas Card, William Agnew, Ravit Dotan, and Michelle Bao. The values encoded in machine learning research. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022.
- [Crawford, 2021] Kate Crawford. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, New Haven, 2021.
- [Headrick, 1991] Daniel R. Headrick. *The Invisible Weapon: Telecommunications and International Politics, 1851-1945*. Oxford University Press, New York, 1991.
- [Hillman, 2021] Jonathan E. Hillman. *The Digital Silk Road: China's Quest to Wire the World and Win the Future*. Harper Business, New York, 2021.
- [Moiloa, 2026] Palesa Moiloa. Silicon valley's vision for global ai is flawed: each country needs its own blueprint. *Nature*, 654:844, 2026.
- [Waight *et al.*, 2026] Hannah Waight, Eddie Yang, Yin Yuan, Solomon Messing, Margaret E. Roberts, Brandon M. Stewart, and Joshua A. Tucker. State media control influences large language models. *Nature*, pages 1–9, 2026.